ThermoBridge

1 / Design choices and delivery status

Unified Engineering Workbench. This project implements the brief directly: one native Windows C++17 interface for geometry imports, a verified 1D thermal solver, multicore experiments, embedded-device simulations and a small backpropagation exercise. The main design choice is to reduce packaging risk and keep engineering behavior inspectable.

Layer	Responsibility
Native Win32 / GDI UI	File dialogs, views, orthographic camera, labelled plots and status. Background tasks preserve responsiveness.
C++ core	Strict import parsing, FE assembly, Thomas solver, benchmark worker pool, explicit gradients and device mocks.
Evidence	Real Linux tests and measurements; C++ CSV exports; source history, dependency notices and operating guide.

Why this scope

A standard-library numerical core avoids commercial solvers and keeps the important logic visible. GDI wireframe rendering is sufficient for STL triangles; it deliberately avoids adding a CAD kernel. The hardware path is simulated so no physical board or paid tooling is needed. Application processing and concurrency are C++17; Python was used only to prepare report graphics from the C++ outputs.

What has and has not been verified

Verified: Linux core build/tests, numerical convergence, central differences, serial/parallel agreement, Windows x64 cross-compilation and PE dependency inspection. **Pending:** actual Windows 11 clean-machine launch, target editor run and real UI walkthrough. A Wine smoke attempt was blocked by server-socket restrictions. This is not a passing standalone-launch claim.

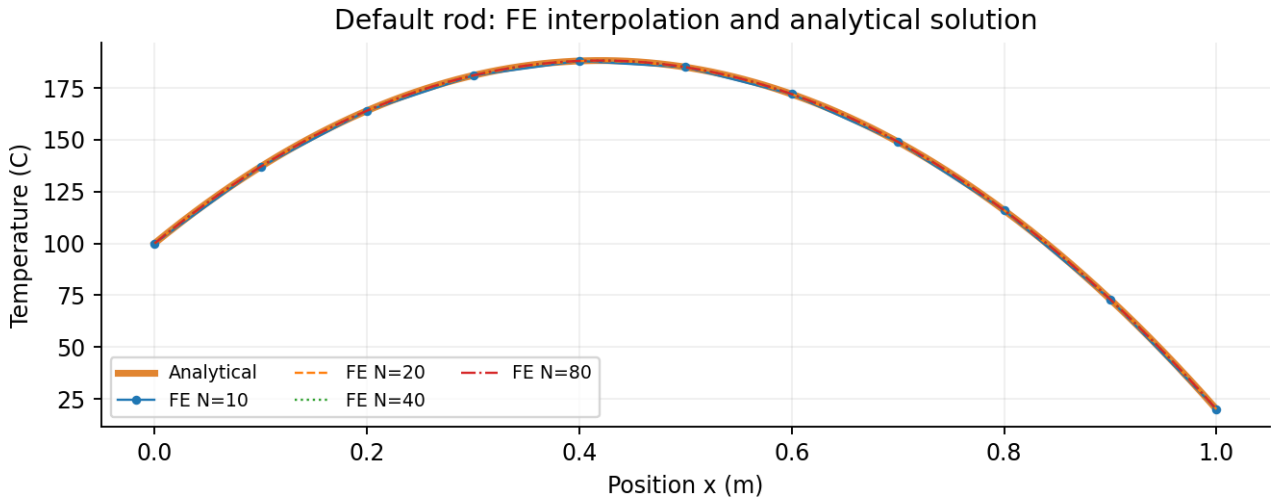
The candidate must also supply their own legible handwritten appendix (maximum two pages), actual effort record, candidate ID and accurate ownership confirmation. The package contains a study derivation and recording plan, not fabricated handwritten pages or screenshots.

Thermal verification

2 / Formulation and analytical comparison

Uniform rod: L=1 m, A=1e-4 m², k=10 W/(m K), q=10,000 W/m³, T(0)=100 C, T(L)=20 C. Constant properties, steady state, insulated sides and one-dimensional axial conduction. Solve -d(kA dT/dx)/dx=qA.

For each element of length h: K_e=(kA/h)[[1,-1],[-1,1]], f_e=(qAh/2)[1,1]. Assembly adds overlapping nodal contributions. Prescribed endpoint columns are eliminated into the RHS before endpoint rows become identity rows. The Thomas algorithm solves the resulting tridiagonal system in O(N) time and memory.



N	Midpoint error (C)	Error reduction
10	1.25000000	-
20	0.31250000	4.000
40	0.07812500	4.000
80	0.01953125	4.000

Analytical T=100-80x/L+[q/(2k)]x(L-x). The FE midpoint temperature is $(T_i+T_{i+1})/2$, not the exact function evaluated there. Maximum error is qh²/(8k), hence second-order convergence. Exported CSVs contain nodes and midpoints.

Debugging insight: nodal-only error is misleading for this constant-property, uniformly forced problem. Coarse-mesh nodal values can already match the quadratic solution closely while interpolation between nodes is still wrong. Midpoint checks expose that error.

Imports and reliability

3 / Data contract, validation and tests

The bundled 12-triangle prism is self-created ASCII STL. Its coordinate units are assumed meters because STL itself has no unit declaration. The 84-node CSV is explicitly **synthetic**, with the required ANSYS-export schema; it is not claimed to come from ANSYS. Both examples are CC0 and have provenance notes.

Feature	Implemented behavior / limit
STL	Rotate, pan, zoom, fit-to-view wireframe triangles. Reject missing tokens, zero-area triangles, non-finite data and trailing content.
Nodal CSV	Coloured 3D points, Celsius legend and nearest-node inspection. Exact five-column header, unique IDs and finite numbers.
Limits	8 MiB/file; 20,000 triangles; 50,000 nodes. Exact triangle/node caps and oversized input are exercised in tests.
Failure handling	Empty/malformed inputs produce actionable messages. Prior valid imports remain available.
Background jobs	One UI worker; independent benchmark workers inside the core. Completed results published only after joining. Cancel discards partial work.
Export	Completed CSV buffer written to a temporary sibling then renamed. On Windows choose a new name if the destination already exists.

Test outcomes

The final core suite contains 57 passing checks: Dirichlet endpoints, midpoint errors, convergence, zero-source linear case, two-element assembly, invalid parameters, good/bad imports, parser boundary sizes, repeatable batches, exact parallel agreement, cancellation, finite-difference gradients and device recovery. AddressSanitizer and UBSan checks pass with leak detection disabled because process-tracing restrictions prevent LeakSanitizer here.

No real CAD solid repair, binary STL, native ANSYS project/results database, mesh-to-result registration or general 3D FEA is implemented. Imported coordinates are rendered; they do not change the separate uniform-rod analysis. Smooth rendering at every size cap on every machine is not claimed.

Evidence: tests/core_tests.cpp; results/linux_run/tests.log and sanitizers.log; data/PROVENANCE.md; src/core/import.cpp. GUI layout and interaction still require a Windows smoke test.

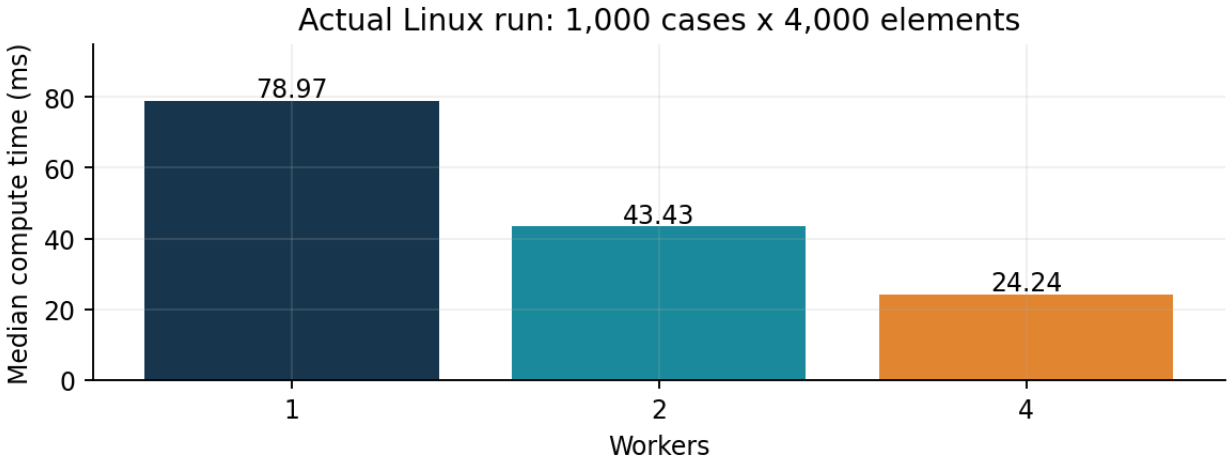
Multicore measurements

4 / Reproducible workloads and honest timing

Actual Linux container run: AMD EPYC 9V74 host model string, 9 logical CPUs and 9 guest-visible cores, 10,451,427,328 host-visible RAM bytes. This is virtualized/shared infrastructure; true host physical topology and resource guarantees are unknown. Compiler GNU 13.3.0, C++17, -O3 -DNDEBUG, pthreads.

1,000 cases x 4,000 elements; mt19937 seed 2027. k varies from 5 to 25 W/(m K), q from 1,000 to 20,000 W/m³. Cases are generated once. All worker counts run the same assembly and solve. A bounded std::thread pool obtains indices atomically and writes each case to its own preallocated result slot.

Workers	Median (s)	Speedup	Efficiency	Max delta (C)
1	0.078966	1.000x	100.0%	0
2	0.043431	1.818x	90.9%	0
4	0.024241	3.258x	81.4%	0

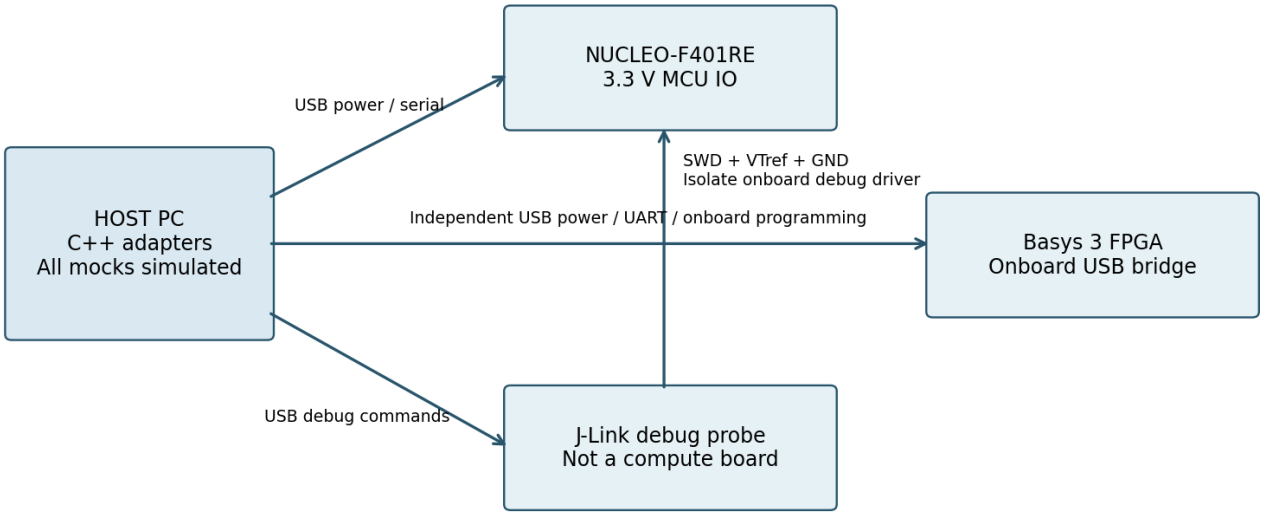


Protocol: a serial reference, one untimed warm-up per worker count, then five measured repetitions. Median reported. steady_clock wall time includes allocation, thread creation, computation and joining; excludes generation, numerical comparison, disk IO, rendering and compilation. All per-repeat temperature comparisons matched exactly (maximum delta 0 C, tolerance 1e-8 C). Raw times are in performance.csv.

Speedup=T1/Tp; efficiency=speedup/p. These short workloads can be affected by shared-host load, allocator/cache effects and thread startup. No minimum or portable speedup is claimed. The GUI can rerun configurable workloads on the candidate machine. Launch-to-ready is separate: the app logs entry-to-ready-timer, but real double-click-to-ready remains unmeasured.

Embedded integration

5 / Distinct voltage domains and device roles



Design-only connections. Verify exact board/probe revisions before wiring. No direct MCU-FPGA cable required.

NUCLEO-F401RE, Basys 3 and a standard 20-pin SEGGER J-Link are proposed design targets. MCU and FPGA connect independently to the host using their documented USB paths. J-Link is a debug probe; it can control/debug a supported target and is not a generic compute board. This design uses its SWD mode for the MCU. No external probe-to-FPGA cable compatibility is assumed.

Path	Voltage / connector assumptions
MCU + FPGA USB	USB board-power inputs; board regulators own internal rails. Expansion IO requires revision-specific checking. No tied power outputs.
Probe to MCU	3.3 V reference; pin 1 VTref, pin 4 ground, pin 7 SWDIO to PA13, pin 9 SWCLK to PA14, pin 15 reset. Verify board header/pads before wiring.
JTAG ownership	Probe drives TCK/TMS/TDI; target drives TDO. One master only. Isolate onboard ST-LINK for external MCU debug. FPGA uses onboard programming path.
Level shifting	No cross-board GPIO in this design. Add direction-appropriate translation if future IO voltages differ; never assume 5 V tolerance.

The mocks expose discovery, capabilities, connection state and telemetry. A one-shot timeout is recoverable on retry and explicitly labelled SIMULATED. No USB traffic, firmware deployment or programming occurs. See Hardware_Design.md for the full interface table, safe sequence and official reference URLs.

References: ST UM1724 and STM32F401RE datasheet; SEGGER J-Link interface/20-pin documentation. Digilent manual URL is supplied for revision verification; automated access returned 403. The proposed hardware design still requires actual board/revision review before physical use.

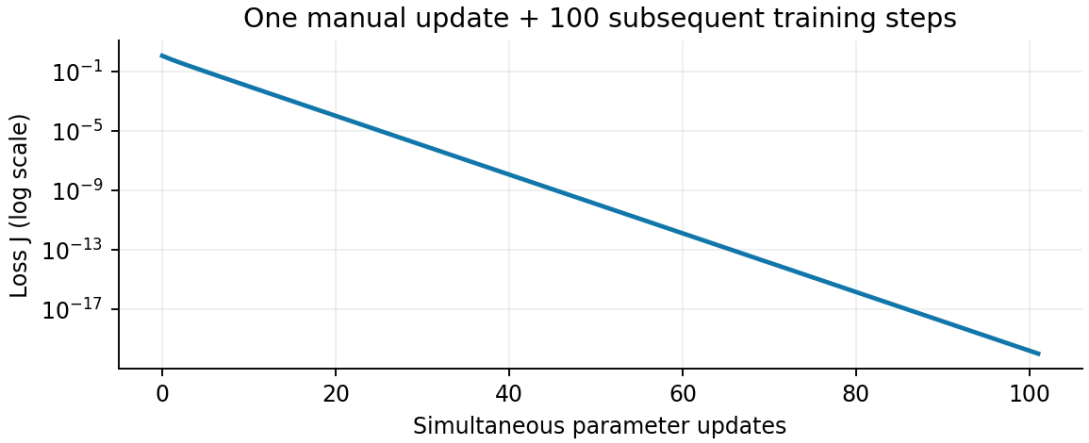
Learning and ownership

6 / Backpropagation, judgement and handover

x=1, y=2; h=tanh(wx+b); prediction=vh+c; J=0.5(prediction-y)². Initial w=0.5, b=0, v=1, c=0. h=0.4621171573, residual=-1.5378828427, J=1.1825418190. Chain rule gives dJ/dv=residual*h, dJ/dc=residual, dJ/dw=residual*v*(1-h²)*x, dJ/db=residual*v*(1-h²).

Parameter	Analytical gradient	FD absolute error
w	-1.2094644752	1.09e-10
b	-1.2094644752	1.09e-10
v	-0.7106820475	5.25e-11
c	-1.5378828427	1.12e-10

With learning rate 0.1, simultaneous update: w=0.6209464475, b=0.1209464475, v=1.0710682047, c=0.1537882843. All gradients use the original parameters. Central finite differences use epsilon=1e-6 and satisfy absolute error <=1e-5.



Loss after the first update is 0.6857737650; after 100 additional updates it is approximately 1.08e-20. Negative gradients mean increasing the parameters locally decreases loss. A single fitted sample says nothing about generalization. The typed derivation is a learning aid; a handwritten scan remains a candidate deliverable.

AI judgement: disclosed AI assistance produced the baseline source and documents. Deliberate challenge tests rejected nodal-only validation and malformed NaN data. Real sequential Git commits and logs are included; no 15-day history, personal effort total or unsupported test result was invented. Candidate review, own edits, handwriting, Windows validation and recorded demonstration are required before submission.